

A Study of Trajectory Planning of Medical Robots Based on Particle Swarm Optimization (PSO) Algorithm

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Abstract—With the continuous popularization of smart healthcare in recent years, medical robots has been a technological representative of smart healthcare, which have been put into wide application with their related technologies. Many researchers have studied and improved them. In this study, the model for a six-axis robotic arm utilized in medical robots was initially constructed using the DH coordinate model, along with its forward and inverse kinematic solutions. Subsequently, the trajectory planning for several predefined surgical actions performed by the medical robots was analyzed. This analysis employed fifth-degree polynomial microelement interpolation and the particle swarm optimization (PSO) algorithm to optimize the trajectory. Finally, simulations were carried out using the MATLAB/Simulink platform to realize the established trajectory movements under the three set medical conditions. Through theoretical modeling and simulation, the feasibility and validity of the theoretical model were verified and provided an effective basis for the design and development of subsequent medical robots.

Keywords—Particle swarm optimization algorithm (PSO), Medical robotics, Six-axis robotic arm, DH modeling, Trajectory planning

I. INTRODUCTION

In recent years, the swift advancement of science and technology has led to a broad array of applications for robots within the realm of intelligent healthcare. Among them, medical robots, as a technological representative of intelligent healthcare, have multiple degrees of freedom in a narrow space making them more flexible, compliant and distal dexterity, and numerous researchers are currently making technological innovations for them [1-3].

Currently, numerous studies on medical robots have been conducted by both domestic and international scholars. Dongeun Choi et al. designed and fabricated a minimally invasive surgical robotic arm and proposed future research directions, including kinematic and kinetic analyses, precise

control, target surgical selection, and design modifications for minimally invasive surgical robotic arms [4]. Zehua Li et al. addressed challenges in minimally invasive surgery, such as limited and variable surgical space and the inflexibility of surgical robots. They designed a small, flexible wire-driven continuous minimally invasive surgical robot and simulated its workspace using the Monte Carlo method. [5]. M.H. Korayem et al. performed experiments with two types of surgical robotic arms and laparoscopic scissors to assess the performance of surgical robots. [6]. Zhengyu Wang et al. explored a feed-forward adaptive PID control method using a nonlinear Luger model and a joint angle estimator. This approach aimed to enhance the dynamic performance and positional tracking accuracy of the robot [7]. Jiahao Qiu et al. introduced an enhanced Monte Carlo sampling technique to significantly boost the sampling rate for computing the workspace of a parallel surgical robot. They achieved precise workspace calculations by encrypting and sampling the workspace boundaries using a boundary sampling method. [8]. Francesco Cursi et al. suggested an adaptive modeling method for developing the forward kinematics of a surgical robot. This approach involved learning a model and integrating it with a tandem jointed robotic arm model to ensure precise task execution, meet additional constraints, and achieve correctness. [9]. Gang Zhang et al. introduced an enhanced Levenberg-Marquardt (ILM) method to model the inverse kinematics of a CCM. They conducted a comparative analysis with a standard continuous robotic arm to support the design of highly dexterous minimally invasive surgical instruments. [10]. Idris Sancaktar et al. explored the use of an adaptive Particle Swarm Optimization (PSO) algorithm for solving the inverse kinematics problem of a medical robot. Their study, conducted on a robot with three linear and three rotational axes, demonstrated improved performance compared to the standard PSO algorithm. [11]. Somyot Kaitwanidvilai and his team have introduced an innovative approach to analyzing and designing robotic systems. The key to the approach is to maximize energy efficiency by

regulating the speed and acceleration of each arm of a six-axis robotic arm to ensure that the robot follows a specific trajectory determined by lean manufacturing methods. [12]

In this paper, firstly, basic modeling was established on the six axis robotic arms used by some medical robots, and DH coordinate models and their forward and inverse solutions were established. Secondly, trajectory planning was carried out for some of the established surgical procedures to be performed by the medical robots, and the trajectory of the medical robots was investigated using the interpolation method. Finally, simulation was carried out using the MATLAB/Simulink platform to realize the established trajectory movements under the three set medical conditions.

II. MODDLING

A. DH Coordinate System Modeling

The common methods for modeling robotic arms are primarily divided into two types: standard D-H parameter modeling and improved D-H parameter modeling. Among them, the standard D-H parameter modeling is more suitable for some medical robotic arms that simulate the human arm configuration.

In this case, the modeling method utilized in this paper is standard D-H parametric modeling, whose parameters are shown in Fig.1.

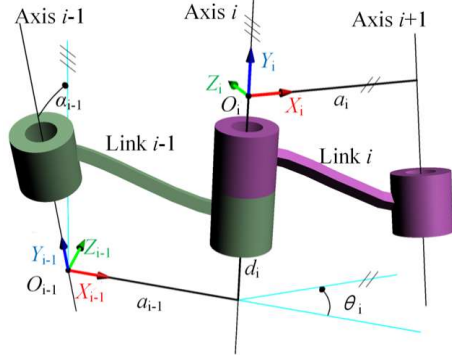


Fig.1. Standard D-H parameter modeling.

The coordinate system is established based on the actual model. Initially, the base coordinate system is determined, followed by defining the Z-axis and X-axis for each joint. Lastly, the Y-axis is determined using the right-hand rule. Where i denotes the joint serial number, θ_i denotes the joint angle θ degree, d_i is the distance between the neighbouring joints cheaply, a_{i-1} denotes the length of the connecting rod of the two adjacent joints, and a_i denotes the joint torsion angle. Where θ , d , a are fixed values, and α is variables, the D-H parameter table is presented in TABLE I.

TABLE I. D-H PARAMETER

Joint i	Connecting rod angle α_i	Rod length a_i	Connecting rod offset d_i	Joint angle θ_i
1	$\pi/2$	0	d_1	θ_1
2	0	$-a_2$	0	θ_2
3	0	$-a_3$	0	θ_3
4	$\pi/2$	0	d_4	θ_4
5	$-\pi/2$	0	d_5	θ_5
6	0	0	d_6	θ_6

B. Kinematics Model Establishment and Solution

The kinematics of the six-degree-of-freedom robotic arm mainly includes forward kinematics and inverse kinematics, the former is to know the desired angle running trajectory information of each joint, and then carry out forward kinematics solution, to get the corresponding state variables of the end of the robotic arm trajectory. The latter is to know the trajectory and then solve the inverse kinematics to calculate the angular state variables of each joint.

1) Positive kinematics solution

The transformation matrix from coordinate system $\{i-1\}$ to coordinate system $\{i\}$ can be expressed using the chain rule for coordinate system transformations as follows:

$${}^{i-1}_iT = {}^{i-1}_R T_Q^R T_P^Q T_i^P T = \text{Rot}(X_{i-1}, \alpha_{i-1}) \text{Trans}(X_{i-1}, a_{i-1}) \text{Rot}(Z_i, \theta_i) \text{Trans}(Z_i, d_i) \quad (1)$$

$$= \begin{bmatrix} \cos \theta_i & -\sin \theta_i & 0 & a_{i-1} \\ \sin \theta_i \cos \alpha_{i-1} & \cos \theta_i \cos \alpha_{i-1} & -\sin \alpha_{i-1} & -d_i \sin \alpha_{i-1} \\ \sin \theta_i \sin \alpha_{i-1} & \cos \theta_i \sin \alpha_{i-1} & \cos \alpha_{i-1} & d_i \cos \alpha_{i-1} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The improved D-H parameters of the six-degree-of-freedom robotic arm for medical use are brought into Eq. (1) to obtain the chi-square transformation matrices between adjacent axes, and all the chi-square transformation matrices are multiplied together:

$${}^0_6T = {}^0_1T \times {}^1_2T \times {}^2_3T \times {}^3_4T \times {}^4_5T \times {}^5_6T \quad (2)$$

The chi-square variation matrix of the end position with respect to the reference coordinate system is obtained:

$${}^0_6T = \begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (3)$$

where p_x, p_y, p_z are the coordinates of the end of the robot arm in the base coordinate system; n_x, n_y, n_z are the direction vectors of the x-axis of the end-of-arm coordinate system in the base coordinate system; o_x, o_y, o_z are the direction vectors of the y-axis of the end-of-arm coordinate system in the base coordinate system; and a_x, a_y, a_z are the direction vectors of the z-axis of the end-of-arm coordinate system in the base coordinate system.

2) Inverse kinematics solution

According to the knowledge related to inverse kinematics, it is only known that the robot's end position matrix is T , and the joint angle in the chi-square transformation matrix of each adjacent coordinate system is also an unknown parameter, so the homogeneous transformation matrix T is expressed by positive kinematics as:

$$\begin{matrix} {}^1T \\ {}^2T \\ {}^3T \\ {}^4T \\ {}^5T \\ {}^6T \end{matrix} \begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = {}^0T = {}^0T \times {}^1T \times {}^2T \times {}^3T \times {}^4T \times {}^5T \times {}^6T \quad (4)$$

According to the principle of positive kinematics, the inverse solution of the matrix of chi-square transformations between coordinates can be found, and θ_1 , θ_5 , θ_6 , θ_3 , θ_2 , θ_4 can be found sequentially through trigonometric functions and equation transformations:

$$\theta_1 = a \tan 2({}^0P_{5y}, {}^0P_{5x}) \pm a \cos\left(\frac{d_4}{\sqrt{{}^0P_{5x}^2 + {}^0P_{5y}^2}}\right) \quad (5)$$

$$\theta_5 = \pm a \cos\left(\frac{{}^0P_{6x} \sin \theta_1 - {}^0P_{6y} \cos \theta_1 - d_4}{d_6}\right) \quad (6)$$

$$\theta_6 = a \tan 2\left(\frac{-{}^6\hat{X}_{0y} \cdot \sin \theta_1 + {}^6\hat{Y}_{0y} \cdot \cos \theta_1}{\sin \theta_5}, \frac{-{}^6\hat{X}_{0x} \cdot \sin \theta_1 + {}^6\hat{Y}_{0x} \cdot \cos \theta_1}{\sin \theta_5}\right) \quad (7)$$

$$\theta_3 = \pm a \cos\left(\frac{|{}^1P_{4xz}|^2 - a_2^2 - a_3^2}{2a_2a_3}\right) \quad (8)$$

$$\theta_2 = \phi_1 - \phi_2 = a \tan 2({}^{-1}P_{4z}, {}^{-1}P_{4x}) - a \sin\left(\frac{-a_3 \sin \theta_3}{|{}^1P_{4xz}|}\right) \quad (9)$$

$$\theta_4 = a \tan 2({}^3\hat{X}_{4y}, {}^3\hat{X}_{0x}) \quad (10)$$

In the inverse solution solving process, θ_1 corresponds to two sets of solutions, so each θ_1 can be solved to one θ_5 , so that θ_5 can get four solutions, each θ_5 corresponds to one θ_6 , and θ_6 therefore also has four sets of solutions; each θ_5 corresponds to two θ_2 , θ_3 , and θ_4 does not appear to have too many solutions, so there are a total of eight sets of inverse kinematics solving of the six-degree-of-freedom robotic arm.

At the same time, in order to make the medical robotic arm work accurately and quickly to complete the target task and reduce energy consumption, it is necessary to select the optimal solution that meets the conditions from multiple groups of inverse solutions, and select the optimal solution according to the principle of the minimum travelling distance, i.e., even if it is possible to get the minimum absolute sum of the transformed amount of the angular value of each joint to be selected.

C. Trajectory Planning

The solution for positive inverse kinematics reveals that the position of a six-degree-of-freedom medical robotic arm in space often corresponds to a specific point. However, in medical scenarios requiring specific actions, the arm's movement typically involves multiple such points. Therefore, to execute the arm's movement in space comprehensively, trajectory planning is necessary, as illustrated in Fig.2.

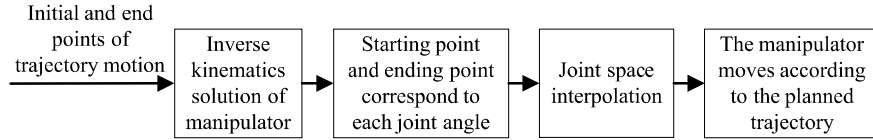


Fig.2. Flow chart of manipulator trajectory planning.

1) Microelement Method

The microelement method is the basis of trajectory planning, which achieves control of the motion of a six-axis robotic arm for medical use by dividing the entire path into multiple discrete points. For a given start point and target point, the path is first divided into multiple segments, and then the motion parameters of each segment, such as joint angles and velocities, are calculated, and the tangent direction and curvature at each discrete point are computed using the method of differential geometry, to determine the attitude and motion state of the robotic arm at each discrete point.

The key to the microelement method is the calculation of the motion parameters at the discrete points. For two adjacent discrete points P_i and P_{i+1} the tangent direction T_i can be calculated by the following equation:

$$T_i = \frac{P_{i+1} - P_i}{\|P_{i+1} - P_i\|} \quad (11)$$

And the curvature K_i can be calculated by the following equation:

$$K_i = \frac{\|T_{i+1} - T_i\|}{\|P_{i+1} - P_i\|} \quad (12)$$

The motion attitude and curvature information of the robotic arm at each discrete point can be obtained by the micro-element method to achieve a smooth motion trajectory.

2) Interpolation method

After obtaining the motion parameters of the discrete points, we need to interpolate according to the parameters and the micro-element method to obtain the continuous motion curve. In this paper, a quintic polynomial interpolation method is used to smoothly interpolate the joint angles between discrete points to ensure that the motion trajectory of the manipulator is continuous and smooth.

Assuming that the joint angles at discrete points P_i and P_{i+1} are given as q_i and q_{i+1} , respectively, the interpolated expression for the joint angle $q(t)$ is obtained by means of a fifth-degree polynomial interpolation equation:

$$q(t) = a_0 + a_1t + a_2t^2 + a_3t^3 + a_4t^4 + a_5t^5 \quad (13)$$

where t is the time variable and $a_0, a_1, a_2, a_3, a_4, a_5$ are interpolation coefficients that can be determined from the position, velocity, and acceleration conditions at the discrete points.

Through the fifth-degree polynomial interpolation, we can obtain continuous joint angle curves. The schematic diagram of the joint angle curve of the robotic arm generated by the microelement and interpolation method, as shown in Fig.3.

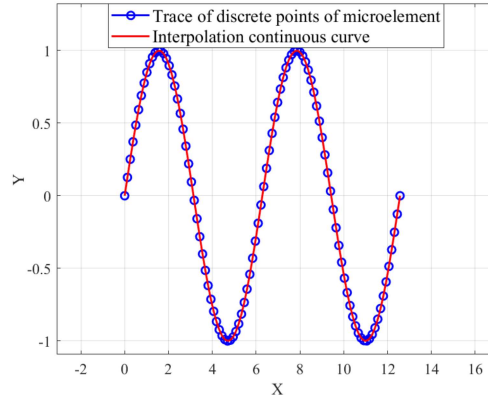


Fig.3. Schematic diagram of discrete point trajectory generated by microelement method.

3) Particle Swarm Optimization Algorithm

In 1995, James Kennedy and Russell Eberhart developed a simplified algorithm model inspired by the foraging behavior of bird flocks. This model was continuously refined, eventually leading to the creation of the particle swarm optimization (PSO) algorithm, also known as the particle swarm algorithm. PSO originated from the observation of how bird flocks share information to collectively find the best location. Imagine a scenario where a group of birds is randomly searching for food in a forest, aiming to locate the spot with the most food. However, each bird doesn't know the exact location but perceives the general direction of the food. Each bird searches in its perceived direction and notes where it finds the most food. They then share this information within the flock, allowing them to collectively identify the location with the most food. During the search, each bird adjusts its search direction based on its past experiences of finding food and the collective information shared by the flock. Over time, the flock converges on the location with the most food, representing the global optimal solution.

In engineering applications, the basic idea is to initialize i particles in a D -dimensional target search space, each particle individually searches for the optimal solution in the search space, and to find the optimal value of the group, all the particles carry out constant iterations, updating their positions and velocities by tracking the individual extremes and the group extremes, which are updated through Eq. (14):

$$v_{id}^{t+1} = \omega v_{id}^t + c_1 r_1 (p_{id}^t - x_{id}^t) + c_2 r_2 (p_{gd}^t - x_{id}^t) \quad (14)$$

$$x_{id}^{t+1} = x_{id}^t + v_{id}^{t+1} \quad (15)$$

where ω is the inertia weight, regulating the search range of the solution space; c_1 and c_2 are the learning factors, regulating the maximum learning step; r_1 and r_2 denote the random numbers on the interval $[0,1]$; t is the number of iterations; v is the iterative speed of the particles; x is the

position of the particles; p_{id}^t is the individual extremum, and p_{gd}^t is the population extremum.

Through Eq. (14), it can be seen that in the process of constant particle iteration, all the particles are updated only in the direction of the individual extremes and group extremes, which leads to the particle population being updated towards the same position in the later iterations, the iteration speed is slow, and it is easy to fall into the local extremes, which makes the particle swarm's global optimization seeking ability decrease.

To optimize the trajectory of the medical robot, the optimization objective is set to minimize the robot time, and at the same time, the energy consumption of the robot during the movement is taken into account while minimizing the energy consumption is considered as a secondary objective of the optimization.

To evaluate the merits of the trajectory, the following fitness function is designed:

$$f(x) = \omega_1 \cdot \text{Time}(x) + \omega_2 \cdot \text{Energy}(x) \quad (16)$$

where x is the joint angle matrix of the robot, $\text{Time}(x)$ denotes the time required for the robot to move to the target position, and $\text{Energy}(x)$ denotes the energy consumed by the robot during the movement process. ω_1 and ω_2 are the weight coefficients to regulate the importance of time and energy in the optimization. A specific flowchart of the particle swarm optimization algorithm for optimizing a specific motion trajectory of a medical robot is shown in Fig.4.

The basic steps are as follows:

- 1) Initialize the population x ;
- 2) Calculate individual fitness;
- 3) Update the particle velocity and position;
- 4) Calculate the fitness of the new position, if the fitness of the new position is higher, then update the position of the

particle, otherwise do not update.

5) Determine whether the termination is satisfied, if yes, exit, otherwise return to the second step.

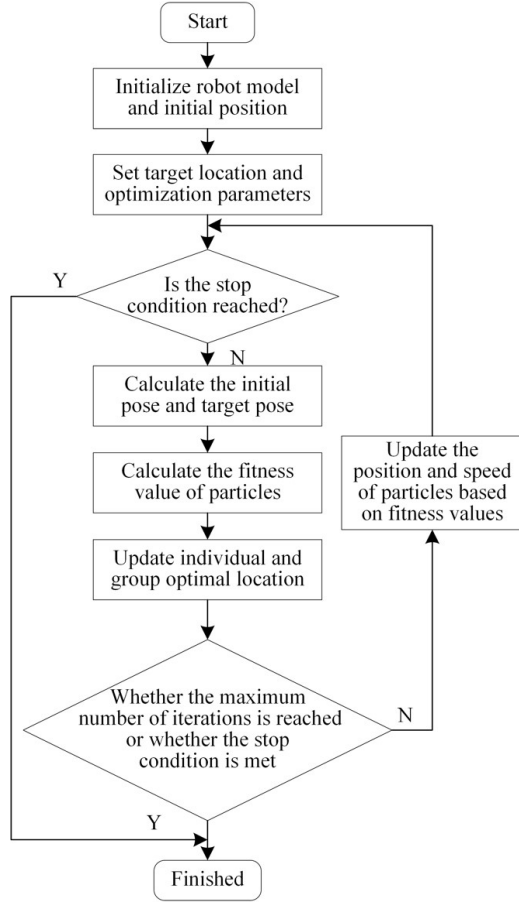


Fig.4. Process of PSO algorithm for optimizing the specific motion trajectory of medical robot.

III. SIMULATION

A. Simulation Model

1) Microelement Method

The mathematical model of the surgical robot was initially developed using the Matlab Robotics Toolbox. The surgical robot uses D-H parameters, as shown in TABLE II.

TABLE II. D-H PARAMETER TABLE OF SURGICAL ROBOT

Joint i	Connecting rod angle α_i	Rod length a_i	Connecting rod offset d_i	Joint angle θ_i
1	$\pi/2$	0.025	0.4	0
2	0	0.56	0	$\pi/2$
3	$\pi/2$	0.035	0	0
4	$\pi/2$	0	0.515	0
5	$\pi/2$	0	0	0
6	0	0	0.08	0

When performing the opening action, the surgical robot first cuts down vertically, defining the scalpel to be 15 cm away from the end of the robot, and after cutting down vertically by 1 cm, it performs the scribing action, ensuring the depth of the incision by rotating the end by 45 degrees, and finally the end returns to the starting position,

completing the opening to action once.

In the above three steps, the start and end points of each step can obtain the robot's start position and end position as well as the corresponding joint angle, the trajectory planning for the intermediate trajectory, the use of fifth-degree polynomials for the microelement processing, the selection of 20 sampling points, the use of interpolation methods to fit the specific trajectory of the surgical robot, to obtain the final trajectory of the trajectory, and run the software in the MATLAB software. After that, the specific trajectory of the surgical robot to perform the opening action can be obtained.

2) Suture robot performing stitching behavior

The mathematical model of the surgical robot was initially developed using the Matlab Robotics Toolbox. The stitching robot uses D-H parameters, as shown in TABLE III.

TABLE III. D-H PARAMETER TABLE OF SUTURE ROBOT

Joint i	Connecting rod angle α_i	Rod length a_i	Connecting rod offset d_i	Joint angle θ_i
1	$\pi/2$	0.025	0.4	0
2	0	0.56	0	$\pi/2$
3	$\pi/2$	0.035	0	0
4	$\pi/2$	0	0.515	0
5	$\pi/2$	0	0	0
6	0	0	0.08	0

In performing the stitching action, the initial position of the end-effector is first set to (0.5, 0.5, 0.15) and the end position to (0.5, 0.5, -0.25). The initial and end positions of the end-effector and the corresponding joint angles are subsequently calculated. After generating the joint angles for the trajectory movement, horizontal movement is added to achieve the stitching action, and the number of steps is generated based on the average of the movement time, with an even number of steps causing the end-effector to move 0.2 radians to the right, and an odd number of steps moving 0.2 radians to the left. Finally, the whole motion trajectory is processed using a fifth-degree polynomial for microelement processing, 20 sampling points are selected, and the interpolation method is used to fit the specific motion trajectory of the sewing robot to obtain its final motion trajectory, which can be obtained after running in the MATLAB software to carry out the sewing needle action of the sewing robot.

3) Medicine pickup robot performs medicine pickup actions

The mathematical model of the surgical robot was initially developed using the Matlab Robotics Toolbox. The surgical robot uses D-H parameters, as shown in TABLE IV.

TABLE IV. D-H PARAMETER TABLE OF MEDICINE TAKING ROBOT

Joint i	Connecting rod angle α_i	Rod length a_i	Connecting rod offset d_i	Joint angle θ_i
1	$\pi/2$	0.025	0.4	0
2	0	0.56	0	$\pi/2$
3	$\pi/2$	0.035	0	0
4	$\pi/2$	0	0.515	0
5	$\pi/2$	0	0	0
6	0	0	0.08	0

When executing the pickup action, the initial position of the end-effector is set to be (0.7, 0.5, 0.15) and the target position is set to be (0.7, -0.5, 0.05). Firstly, the initial orientation of the end-effector and the target position need to be calculated. Then, the joint angles of the end-effector at both the initial and target positions are computed. During trajectory planning, a fifth-degree polynomial is utilized for microelement processing, with 20 sampling points selected. Interpolation is used to fit the motion trajectory of the drug-taking robot based on these points. Through simulation in MATLAB software, the specific trajectory for the drug-taking action of the robot can be obtained.

B. Analysis of Results

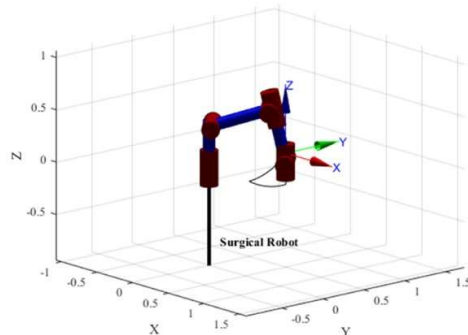


Fig.5. Simulation diagram of surgical robot performing operation.

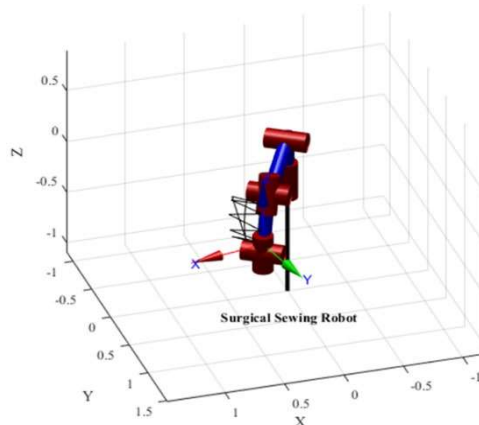


Fig.6. Simulation diagram of sewing robot executing sewing action.

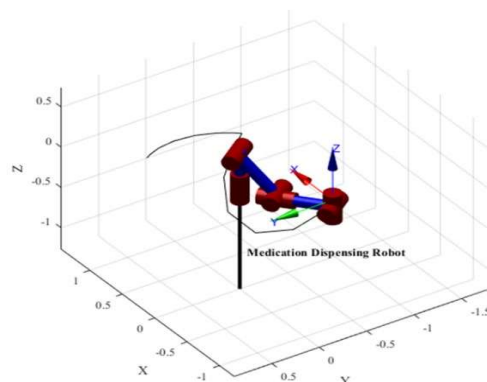


Fig.7. Simulation diagram of drug taking robot executing drug taking action.

Relying on MATLAB software for simulation, the

surgical robot executing the opening action as shown in Fig.5, the suturing robot executing the suture action as shown in Fig.6, and the medicine picking up robot picking up the medicine action as shown in Fig.7 were realized respectively. According to the simulation experiments, the medical robots run smoothly and accurately in all stages when performing different specific tasks, the action speed is fast and smooth, and there is no jitter and other phenomena, which verifies the validity and practicality of the established motion planning.

IV. CONCLUSION

In this paper, firstly, the basic modeling of the six-axis robotic arm used in some medical robots was carried out, and the D-H coordinate model and its forward and inverse solutions were established. Secondly, trajectory planning was investigated using fifth degree polynomial microelements and interpolation for some of the surgical established actions that need to be performed by the medical robots, and particle swarm optimization algorithm was introduced to optimize the established trajectories. Finally, simulations were carried out using the MATLAB--Simulink platform to achieve the established trajectory movements under the three set medical conditions, and the feasibility and validity of the theoretical model were verified through theoretical modeling and simulation, and an effective basis was provided for the design and development of subsequent medical robots.

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